

MILAN SINGH

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SUMMARY

Full-Stack and React Native Developer who ships cross-platform mobile apps and scalable web backends end-to-end. Strong with React, Next.js, React Native, Node.js, Express, Fastify, PostgreSQL, and MongoDB, including real-time data, push notifications, auth, and AWS deployment.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, SQL, HTML, CSS

Frontend: React, Next.js, React Native (Expo), Redux Toolkit, Zustand, TanStack Query & Router, React Hook Form, Zod, Tailwind CSS, shadcn/ui

Backend: Node.js, Express.js, Fastify, REST APIs, WebSockets, MQTT, JWT & cookie auth, Clerk, Better Auth

Databases: PostgreSQL, MongoDB (Mongoose, aggregation pipelines), Redis, Appwrite

Testing & DevOps: Jest, AWS (Lambda, EC2, S3, API Gateway, Aurora, CloudFront, CloudWatch, IAM), Docker, GitHub Actions, CI/CD, Linux

Concepts: System Design, Microservices, Caching, Rate Limiting, BLE, OTA Updates, Push Notifications

PROFESSIONAL EXPERIENCE

Software Developer · Dash Dynamic *Sept 2025 - Present*

IIT Delhi, India

- Built a cross-platform React Native (Expo) app with high-frequency BLE data streaming to ESP32 devices, including auto-reconnection, failover, offline support, and secure authentication.
- Implemented an end-to-end OTA firmware update pipeline with SHA-256 binary verification on the mobile client and a controlled release lifecycle to prevent premature device updates.
- Engineered a production push-notification system (BullMQ queues, chunked sends, receipt-stage dead-token pruning) and OTP auth with SMS auto-read and rate limiting.
- Designed Fastify backend services over PostgreSQL with optimized queries and Redis caching; built MQTT telemetry pipelines where chargers continuously report availability, with load balancers handling failover to keep the system responsive under load.
- Built a data-intensive operations dashboard with real-time charting, interactive Leaflet maps with OSRM routing, client-side caching, and role-based access control. Deployed on fault-tolerant AWS infrastructure.

Frontend Developer · Sudo Yantra *Nov 2024 - Feb 2025*

New Delhi, India

- Cut initial load time substantially through code splitting, lazy loading, and asset optimization within an Agile team.
- Raised accessibility to WCAG AA (semantic HTML, ARIA, keyboard navigation) and reduced image bandwidth ~40% via WebP and responsive images, improving Core Web Vitals.

PROJECTS

BingeBee — Video Streaming Platform

React, TypeScript, Express.js, MongoDB, Redis, TanStack · [bingebbee.vercel.app](#) · [github.com/milan-sh/bingebbee](#)

- Built a full-stack YouTube-style platform with public browsing, video upload via Multer + Cloudinary, watch history, playlists, subscriptions, likes, comments, community posts, and user profiles.
- Wrote a custom Express backend with JWT + httpOnly cookie auth and complex MongoDB aggregation pipelines for feeds, subscription, subscribers, and watch history.
- Implemented type-safe forms and data fetching with TanStack Query & Router, React Hook Form, and Zod, plus client-side caching for fast navigation.

Happy Dates — Birthday Reminder App

React Native, Express.js, Better Auth, OpenAI, TanStack, Redis

- Cross-platform app to track birthdays with reminders, push notifications, and AI-generated celebration ideas and personalized messages.

ACHIEVEMENTS

- Smart India Hackathon 2022 — First Runner-Up:** Built a real-time dashboard with sensor integration and data visualization for driver-health monitoring to help prevent heavy-vehicle accidents.

EDUCATION

B.Sc. in Electronic Science · University of Delhi *Jul 2021 - Jun 2024*